

Q.No:9

Problem Description

There are N chocolates denoted by array A where $A[i]$ is the length of the i -th chocolate.

Alice can melt each chocolate and then convert it into a chocolate whose length is any divisor of the number $A[i]$.

So, a chocolate of length $A[i]$ can be converted into X different types of chocolate where X is the count of divisors of the number $A[i]$.

So you need to count the total unordered pair of chocolates such that their X value is same.

Constraints

$1 \leq N \leq 10^5$

$1 \leq A[i] \leq 10^6$

Input Format

The first line contains an integer N as input denoting the total number of elements in the array A .

The next line contains N space-separated integers that denote the elements of the array A .

Output Format

In the output, print the total number of ways as mentioned in the statement.

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
2 3 4

EXPECTED OUTPUT

1

Test Case 2

INPUT (STDIN)

10
2 3 4 2 7 6 8 6 4 9

EXPECTED OUTPUT

12

Mandatory Test Cases

Test Case 1

KEYWORD

while(--N)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

8

Test Case 2

TOKEN COUNT

345

Test Case 3

NLOC

44

Code:

```
#include <stdio.h>

int divisors(int n)
{
    int count = 0;

    for (int i = 1; i * i <= n; i++)
    {
        if (n % i == 0)
        {
            count++;
        }
    }
}
```

```
if (i != n / i)
count++;
}
}

return count;
}

int main()
{
int N;
scanf("%d", &N);

int a[N];
int d[N];

for (int i = 0; i < N; i++)
{
scanf("%d", &a[i]);
d[i] = divisors(a[i]);
}

long long answer = 0;

int i = 0;

printf("Output\n");
while(--N >= 0)
{
int j = i + 1;

while (j < i + 1 + N)
{
if (d[i] == d[j])
{
answer++;
}

j++;
}

i++;
}

printf("%lld\n", answer);

return 0;
}
```

Output:

Test_case: 1

```
case 7 graph TD
  3
  2 3 4
  Output
  1
aadityajaiswal@Aad
```

Test_case: 2

```
case 7 graph TD
  10
  2 3 4 2 7 6 8 6 4 9
  Output
  12
aadityajaiswal@Aadityas-MacBook-Air
```